

Kelly M. Diamond

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Professional Appointments

- 2022-Present Assistant Professor, Rhodes College, Memphis, TN
- 2020-2022 Postdoctoral fellow, Seattle Children's Research Institute, Seattle, WA

Education

- 2019 Ph.D. Biological Sciences, Clemson University, Clemson, SC
Functional morphology and environmental impacts on migration-related performance in amphidromous goby fishes
- 2015 M.S. Biological Sciences, Clemson University, Clemson, SC
*Environmental effects on fish escape responses: impact of flow on the escape performance of the Hawaiian stream goby, *Sicyopterus stimpsoni**
- 2013 B.S. Biology, University of Central Florida, Orlando, FL
Phylogenetic relationships and morphological variation in pitviper taxa

Peer Reviewed Publications

ORCID: 0000-0001-8639-6795 *Undergraduate coauthor, †International Collaborator

23. Schoenfuss HL, **Diamond KM**, Lagarde R[‡], Blob RW. 2026. Ontogenetic change in performance: do innovations constrain performance? *Integrative Organismal Biology*. 8: obag044.
<https://doi.org/10.1093/iob/obag044>
22. **Diamond KM**, Hilliard VK. Form and Function. 2025. In *Comparative Vertebrate and Human Anatomy: Ecology, Evolution, and Function*. Eds. Hilliard VK, Whitenack LB, Ryerson B. PALNI Open Press.
<https://doi.org/10.59319/YHFP2525>
21. **Diamond KM**, Olson C, Utsumi K, Eifler MA, Eifler DA. 2024. Differences between juveniles and adults in habitat use, sprint performance, and morphology in the desert horned lizard, *Phrynosoma platyrhinos*. *Ichthyology and Herpetology*. 112(3): 347-352. <https://doi.org/10.1643/h2023042>
20. **Diamond KM**, Nishiura L, Sakihara T, Schoenfuss HL, Blob RW. 2024. When to go against the flow: examining patterns of performance over multi-day migration events in the Hawaiian stream fish, 'o'opu nōpili (*Sicyopterus stimpsoni*). *Integrative and Comparative Biology*. 64(2): 496-505.
<https://doi.org/10.1093/icb/icae082>
19. Rogers LS, Lozier NR, Sapozhnikova YP **Diamond KM**, Davis JL Sisneros JA. 2023. Functional plasticity of the swim bladder as an acoustic organ for communication in a vocal fish. *Proceedings of the Royal Society B*. 290:20231839. <http://doi.org/10.1098/rspb.2023.1839>

18. McAlpine-Bellis EA*, Utsumi KL*, Klein JRV*, **Diamond KM**, Gilbert-Smith S*, Garrison GE, Eifler M, Eifler D. 2023. Movement patterns and habitat use for sympatric species *Gambelia wislizenii* and *Aspidoscelis tigris*. *Ecology & Evolution*. 13: e10422. <https://doi.org/10.1002/ece3.10422>
17. Munteanu VD, **Diamond KM**, Blob, RW. 2023. Limb bone strains during climbing in green iguanas (*Iguana iguana*): testing biomechanical release as a mechanism promoting morphological transitions in arboreal vertebrates. *Journal of Experimental Biology*. 226: jeb245175. <https://doi.org/10.1242/jeb.245175>
16. Blob RW, **Diamond KM**, Lagarde R, Maie T, Moody KN, Palecek-McClung AM, Ward JL, Schoenfuss HL. 2023. Integrating biomechanics in evolutionary studies, with examples from the amphidromous goby model system. *Journal of Experimental Biology*. 226: jeb244942. <https://doi.org/10.1242/jeb.244942>
15. **Diamond KM**, Burtner AE*, Siddiqui D*, Alvarado K*, Leake SU*, Rolfe S, Zhang C, Kwon RY, Maga AM. 2023. Examining craniofacial variation among crispant and mutant zebrafish models of human skeletal diseases. *Journal of Anatomy*. 243: 66-77. <https://doi.org/10.1111/joa.13847>
14. **Diamond KM**, Good CJ*, Johnny N, Sakihara TS, Edmison PL, Faust JA, Schoenfuss TA, Rubin AM*, Blob RW, Schoenfuss HL. 2022. Assessing Occurrence and Biological Consequences of Contaminants of Emerging Concern on Oceanic Islands. *Water*. 14(3):275. <https://doi.org/10.3390/w14030275>
13. **Diamond KM**, Rolfe SM, Kwon RY, Maga AM. 2022. Computational anatomy and geometric shape analysis enables analysis of complex craniofacial phenotypes in zebrafish mutants. *Biology Open*. 11(2): bio058948 <https://doi.org/10.1242/bio.058948>
12. Elhamod M, **Diamond KM**, Maga AM, Bakis Y, Bart HL, Mabee P, Dahdul W, Leipzig J, Greenberg J, Avants B, Karpatsne A. 2021. Hierarchy-guided neural networks for species classification. *Methods in Ecology and Evolution*. 13: 642– 652. <https://doi.org/10.1111/2041-210X.13768>
11. Rolfe S, Pieper S, Porto A, **Diamond KM**, Winchester J, Shan S, Kirveslahti H, Doug Boyer D, Summers A, Maga AM. 2021. SlicerMorph: An open and extensible platform to retrieve, visualize and analyze 3D morphology. *Methods in Ecology and Evolution*. 12: 1816– 1825. <https://doi.org/10.1111/2041-210X.13669>
10. **Diamond KM**, Lagarde R[‡], Griner JG*, Powder KE, Schoenfuss HL, Walker JA, Ponton D[‡], Blob RW. 2021. Interactions among multiple selective pressures decouple the form-function relationship in stream fishes. *Biological Journal of the Linnean Society*. 134(3): 557–567 <https://doi.org/10.1093/biolinnean/blab098>
9. Forker GK*, Schoenfuss HL, Blob RW, **Diamond KM**. 2021. Bendy to the bone: links between vertebral morphology and waterfall climbing in amphidromous gobioid fishes. *Journal of Anatomy*. 239: 747-754. <https://doi.org/10.1111/joa.13449>
8. Schneider NG*, McDamy AJ*, Rubin AM*, Schoenfuss HL, Blob RW, **Diamond KM**. 2021. Do predators take advantage of prey blind spots? In-stream analysis of predator-prey interactions in Hawaiian stream fishes. *Ichthyological Explorations of Freshwaters*. 30(4): 289-297. <http://doi.org/10.23788/IEF-1162>
7. Leipzig J, Bakis Y, Wang X, Elhamod M, **Diamond KM**, Dahdul W, Karpatsne A, Maga AM, Mabee P, Bart HL, Greenberg J. 2021. Biodiversity image quality metadata augments convolutional neural network classification of fish species. In: Garoufallou E., Ovalle-Perandones MA. (eds) *Metadata and Semantic Research. MTSR 2020. Communications in Computer and Information Science*, vol 1355. https://doi.org/10.1007/978-3-030-71903-6_1

6. Blob RW, Baumann T, **Diamond KM**, Young VKH, Schoenfuss HL. 2020. Functional correlations of axial muscle fiber type proportions in the waterfall-climbing Hawaiian stream fish *Sicyopterus stimpsoni*. *Journal of Anatomy*. 236: 1160–1166. <https://doi.org/10.1111/taja.13169>
5. **Diamond KM**, Lagarde R[‡], Schoenfuss HL, Walker JA, Ponton D[‡], Blob RW. 2019. Ontogenetic change in escape behavior and performance of fishes experiencing different predator regimes. *Biological Journal of the Linnean Society*. 127: 324-336. <https://doi.org/10.1093/biolinnean/blz055>
4. Blob RW, Lagarde R[‡], **Diamond KM**, Keeffe RM*, Bertram RS, Ponton D[‡], Schoenfuss HL. 2019. Functional diversity of evolutionary innovations: insights from waterfall-climbing kinematics and performance of juvenile gobiid fishes. *Integrative Organismal Biology*. Obz029. <https://doi.org/10.1093/iob/obz029>
3. Taft NK, Taft BN, Henck H, **Diamond KM**, Schoenfuss HL, Blob RW. 2017. Comparative morphology and mechanical properties of the lepidotrichia of climbing and non-climbing Hawaiian gobioid fishes. *Cybium*. 41(2): 107-115.
2. **Diamond KM**, Schoenfuss HL, Walker JA, Blob RW. 2016. Flowing water affects fish fast-starts: escape performance of the Hawaiian stream goby, *Sicyopterus stimpsoni*. *Journal of Experimental Biology*. 219: 3100-3105. <http://jeb.biologists.org/content/219/19/3100>
1. **Diamond KM**, Trovillion D, Allen KE, Malela KM, Noble DA, Powell R, Eifler D, Gifford ME. 2014. Individual (co)variation in locomotor performance and field behavior in curly-tailed Lizards, *Leiocephalus carinatus*. *Journal of Zoology*. 294: 248-254. <http://onlinelibrary.wiley.com/doi/10.1111/jzo.12175/full>

Publications in progress

*Undergraduate coauthor, [‡]International Collaborator

1. Avilés-Rodríguez KJ, Lafloe BH, **Diamond KM**, Powell R, Gifford ME, Eifler DA. Search behavior, endurance and morphology in *Pholidoscelis auberi* (Squamata: Teiidae). In revision with the *Journal of Herpetology*.
2. **Diamond KM**, Lagarde R[‡], Maie T, Palecek-McClung AM, Ponton D[‡], Ravichandran S*, Schoenfuss HL, Blob RW. How fixed are fast-starts? Comparative analysis of escape responses in goby fishes from around the world. In preparation.

Grants, Fellowships, and Awards

2024	Rhodes Campus Life Accessibility & Inclusivity Award
2026	Rhodes Hill Grant for Curricular Development and Pedagogical Innovation
2023-2026	Rhodes independent research fellowship for student summer stipends - Using Machine Learning to process Data for Biological Questions (\$18,000)
2019	Vision Research Corporation, Phantom Camera Education Challenge – Awarded equipment usage for filming high speed videos of fish escape responses and climbing trials (eq. \$11,340)

- 2019 Clemson Professional Enrichment Grant (\$1000)
 Clemson Professional Development Grant for Graduate Student Research (\$1000)
 Biological Sciences Graduate Student Association, Travel Award (\$300)
 Alfred 'Hap' Wheeler Distinguished BSGSA Member Award
- 2018 American Philosophical Society, Lewis & Clark Fund for Exploration & Research (\$4300)
 American Museum of Natural History, Theodore Roosevelt Memorial Fund fellowship (\$3413)
 Findley Student Assistance Endowment (\$1500)
 Clemson Professional Enrichment Grant (\$648)
- 2017 Society for Integrative and Comparative Biology Fellowship of Graduate Student Travel (\$2000)
 Biological Sciences Graduate Student Association, Travel Award (\$400)
 Clemson Professional Enrichment Grant (\$750)
 Biological Sciences Graduate Student Association Commitment to Service
- 2016 American Society of Ichthyologists and Herpetologists, Edward C. Raney Grant (\$800)
 Biological Sciences Graduate Student Association, Travel Award (\$400)
 Clemson Professional Enrichment Grants (\$750, \$302, \$414)
- 2015 Biological Sciences Graduate Student Association, Travel Award (\$300)
- 2014 Sigma Xi Grant-in-Aid of Research (\$1000)
 Clemson Professional Enrichment Grant (\$650)
 Biological Sciences Graduate Student Association Commitment to Research
- 2012 Student Government Association Travel Award (\$300)
- 2011 Innovation through Institutional Integration Fellowship (2011-2013)
- 2010 Research and Mentoring Program Scholarship (2010-2011)
 Work Force Central Florida Grant for Undergraduate Research
- 2009 Florida Bright Futures Scholarship (2009-2013)

Teaching and Leadership Experience

- 2025 - present Rhodes Yoga Club Faculty sponsor
- 2023 - present Rhodes Climbing Club - Rockin' Rhodes Faculty sponsor
- 2023 Co-organizer for [ImageDatapalooza](#) held August 14-17, 2023 at Ohio State University
 Hackathon style workshop sponsored by the [Imageomics Institute](#)

- 2023 SEED program mentor
Mentored 2 high school students on an independent research project
- 2022 - present Instructor of record, Rhodes College
Courses: comparative vertebrate morphology, biology 2, research, senior seminar
- 2020-Present Advisory Board Member, Erell Institute
Advise on projects and future directions for the non-profit research institute
- 2021-2022 Research mentor at Seattle Children's Research Institute
Trained and oversaw independent projects of 4 undergraduate students
- 2020-2022 [SlicerMorph](#) Workshop Instructor
Introduction to using SlicerMorph for collecting morphological measurements
- 2018-2019 Founding member of Clemson University, College of Science Outreach Team
Organized science-themed community events
- 2017-2020 Mentor, Erell Institute
Develop research projects with pre-graduate students
- 2016-2019 Mentor, New Foundations foster home for children
Worked one on one with a high school student to encourage learning and self esteem
- 2014-2019 Creative Inquiry Mentor, Clemson University
Trained and oversaw independent projects of 9 undergraduate students
- 2013-2019 Graduate teaching assistant. Clemson University
Labs: Vertebrate Biology, Comparative Vertebrate Morphology, Human Physiology, Mammalogy
- 2016-2018 Lead teaching assistant. Vertebrate Biology Laboratory. Clemson University
Organized 9-12 laboratory sections. Updated content to reflect new findings in the field and emphasize the connection between field observation and course content.
- 2016-2018 Representative for Department of Biological Sciences to the College of Engineering and Sciences Student Advisory Board
PR committee and Outreach committee
- 2014-2018 Guest lecture. Clemson University. Vertebrate Biology
Topics including: Endothermy vs Ectothermy, Bird diversity, Dinosaurs & other Mesozoic diapsids, Mammal diversity
- 2014-2016 Clemson University Natural History Museum. Laboratory and Website Development
Developed content and created interactive displays
- 2011-2013 Mentor, Internet Science and Technology Fair
Worked with e-learning students on science fair projects

Service & Outreach Activities

Professional service

- Reviewer for: *Biodiversitas Journal of Biological Diversity, Biology Letters, Ecology and Evolution, Evolution and Development, Frontiers in Physiology, Ichthyology and Herpetology, Integrative and Comparative Biology, Integrative Organismal Biology, JEZ Part B: Molecular and Developmental Evolution, Journal of Anatomy, Journal of Experimental Biology, Journal of Fish Biology, Methods in Ecology and Evolution, Plos one, Proceedings of the Royal Society B, Royal Society Interface, Scientific Reports*
- 2026 - present Society for Integrative and Comparative Biology (SICB) divisional program officer
Division of Vertebrate Morphology
- 2023 - present SICB Student support committee member
- 2023-2024 Served on the committee for the Society for Integrative and Comparative Biology, Division of Vertebrate Morphology Mentorship Award
- 2021-2026 Assistant Editor for *Integrative and Comparative Biology* journal
Division of Ecology and Evolutionary Biology
- 2018 Panelist, 'Writing a competitive travel grant,' SICB annual meeting, San Francisco, CA
- 2018 Organizer, Graduate Research Brown Bag seminar series
Invited, advertised, and moderated semester long seminar series to highlight graduate student research

Rhodes College Service

- 2026- present Center for Liberal Arts and AI Engagement Fellow for Rhodes College
- 2025 - present Rhodes AI taskforce member
- 2025 Hosted biology department seminar speaker Dr. Anne Sabol from the Memphis Zoo
- 2025 Rhodes AI Summit faculty/staff round table participant
- 2024 - present Rhodes Biology Student Learning Outcomes (SLO3) committee
- 2024 - present Rhodes liaison for the Memphis Zoo
- 2024 Hosted biology department seminar speakers from the NYIT Animal Motion Lab
- 2023 Hosted biology department seminar speaker Dr. Doug Eifler from the Erell Institute
- 2022 - present Graduate school application workshop organizer
Fall 2022; Spring 2023; Spring 2024; Spring 2025
- 2022 Hosted biology department seminar speaker Dr. Rick Blob from Clemson University
- 2022- Present Host prospective Rhodes students in course lectures including BIOL 140/350

Outreach Activities

- 2020-2023 Ask an animal scientist! (<http://askananimalscientist.com/>)
Created a website aimed at encouraging kids of all ages to ask questions about animals during the Covid-19 shutdown
- 2019-2022 Skype a Scientist volunteer
Talk virtually with kids of all ages about biomechanics, ecology, and how science works! I have spoken with students at all age levels in CT, FL, GA, MA, NC, NY, UT, VA, WA, Canada, and the UK through this program!
- 2019-2022 Outreach associate for the journal *Integrative Organismal Biology*.
Wrote blog posts to promote journal articles, see full list here:
<https://iobopen.com/author/kmdiamo/>
- 2017-2018 Clemson Hope- Paw Pals
Helped elementary school students to improve writing skills
- 2017 Tiger Talks
Encouraged elementary school students to pursue their interests in STEM
- 2015-2016 Bring Your Daughter to Clemson Day
Developed interactive, hands-on displays for girls interested in STEM
- 2013-2015 Graduate School Application Workshop, Clemson University
Collected and presented content to help undergraduates apply to graduate programs

Affiliations

- 2019-Present Sigma Xi
Full member since 2023
- 2015-Present International Society of Vertebrate Morphology
- 2013-Present Society for Integrative and Comparative Biology
Divisions of Vertebrate Morphology, Comparative Biomechanics, and Ecology and Evolution

Presentations

Invited Presentations

- 2026 Comparative Jumping Performance of the Critically Endangered Dusky Gopher Frog (*Lithobates sevosus*). **KM Diamond**, C Alton, P Chittoori*, C Hanks, A Sabol, J Swartwood, RH Yang*, S Poo.
Joint meeting of Ichthyologists & Herpetologists, New Orleans, LA.

Using machine learning methods to quantify phenotypes. Rhodes College, Memphis, TN
- 2025 Using math to understand why animals come in so many shapes and sizes and how animals move! Overton Park Science Café, Memphis, TN

Using machine learning methods to quantify phenotype. Arkansas State University. Jonesboro, AR

- Functional anatomy of muscles: Texas A&M Corpus Christi Functional Anatomy course. Virtual
- Why do I have to learn this? How we use math in biology. Darwin Day Talk for the Muddy Waters Free Thought Society. Memphis Museum of Science and History. Memphis, TN
- 2025 Planning semester-long unessay projects in STEM, Rhodes Faculty Lair. Virtual
- 2024 Functional anatomy of muscles: Texas A&M Corpus Christi Functional Anatomy course. Virtual
- When to go against the flow: lessons from migrating fish. **KM Diamond**, HL Schoenfuss, RW Blob. Annual SICB meeting, Seattle WA
- 2023 Functional anatomy of muscles: Texas A&M Corpus Christi Functional Anatomy course. Virtual
- 2022 Using 3DSlicer for SciComm: Tell a story with digital dissections. Society for Integrative and Comparative Biology Workshop. Virtual
- 2021 Using AI to quantify phenotype. Presented virtually at Florida Museum of Natural History and in person at Rhodes College, Memphis, TN
- Becoming an integrative biologist, how to work with people and methods from different fields. St. Cloud State University. Virtual
- Computational anatomy and geometric shape analysis enables analysis of complex craniofacial phenotypes in zebrafish mutants. George Washington University. Virtual
- Using microCT to quantify phenotype, Florida Atlantic University High School Lab Skills Workshop. Virtual
- Ask a Scientist! King County Public Library. Virtual
- 2020 Using machine learning and computational anatomy to quantify phenotype. Seattle Children's Research Institute. Virtual
- Using Slicermorph in Biomechanics. St. Mary's College. Virtual
- Using quantitative anatomy to define cranial phenotypes in mutant zebrafish. University of Washington. Virtual
- Ask an animal scientist live! Bill & Melinda Gates Foundation. Virtual
- 2019 Migration performance: Gobies do more than just climb waterfalls. Wake Forest University, Salem, NC

Student Presentations

- 2025 Comparing canine tooth morphologies across domesticated dog breeds. HH Brown. Presented at Rhodes Symposium, Memphis, TN
- Segmenting super sniffers: a method of digital segmentation of canid nasal structures for morphological analysis. L Martinez. Presented at Rhodes Symposium, Memphis, TN
- 2024 3D modeling protocol for analysis of canine skull morphology: a standardized approach using 3D Slicer. S Venn & S Kessler. Presented at Rhodes Symposium, Memphis, TN

- 2023 The effects of nitrite on Endler's guppies. G Wheeling & B Leake. Presented at Rhodes SURCAS, at the Memphis Summer Scholars Research Symposium, and at the American Chemistry Society Fall meeting in San Francisco, CA

Conference Presentations

*Undergraduate student, **Pre-college Student, †International Collaborator

SICB = Society for Integrative and Comparative Biology

- 2026 Personality in a soft-release study of the critically endangered dusky gopher frog (*Lithobates sevosus*). A Sabol, C Alton, C Hanks, HR Yang*, S Pepperman, **KM Diamond**, S Poo. Joint meeting of Ichthyologists & Herpetologists, New Orleans, LA.

Effects of soft-release on growth, locomotion, and corticosterone in a critically endangered, captive-bred frog. C Alton, C Hanks, **KM Diamond**, L Neuman-Lee, S Pepperman, A Sabol, J Terry, S Poo. Joint meeting of Ichthyologists & Herpetologists, New Orleans, LA.

Identifying patterns of shape convergence in brachycephalic domestic dog breeds. SS Venn, N Hebdon, **KM Diamond**. Annual SICB meeting, Portland, OR.

Comparative analysis of escape responses in goby fishes from around the world. **KM Dimond**, R Lagarde†, T Maie, AM Palecek-McClung, D Ponton†, S Ravichandran, HL Schoenfuss, J Walker, RW Blob. Annual SICB meeting, Portland, OR.

A kinematic comparison of enrichment-based feeding in captive Komodo dragons. G Wheeling*, M Richter, **KM Diamond**. Annual SICB meeting, Portland, OR.

Preferences in enrichment of the Okapi (*Okapia johnstoni*). Z Sarwar*, **KM Diamond**. Annual SICB meeting, Portland, OR.

Beyond the enclosure: kinematics of captive vs. wild meerkats. S Heckmann*, **KM Diamond**. Annual SICB meeting, Portland, OR.

Segmenting super sniffers: A comparative study of nasal turbinal morphology in domestic dog breeds. L Martinez*, N Hebdon, **KM Diamond**. Annual SICB meeting, Portland, OR.

Comparing canine tooth morphology across domesticated dog breeds. HH Brown*, N Hebdon, **KM Diamond**. Annual SICB meeting, Portland, OR.

- 2025 Neurocranial variation in canids: insights on artificial selection from geometric morphometrics. SS Venn*, S Kessler*, N Hebdon, **KM Diamond**. Presented at the Annual SICB meeting, Atlanta, GA and Presented at Rhodes Symposium, Memphis, TN

The impact of zoo visitor presence on meerkat social behavior. SE Heckmann*, **KM Diamond**. Presented at the Annual SICB meeting, Atlanta, GA and Presented at Rhodes Symposium, Memphis, TN

Influence of visitor numbers and noise levels on the behavior of captive meerkats. M Ibrahim*, **KM Diamond**. Annual SICB meeting, Atlanta, GA

- Behavioral differences in two subspecies of Red Panda (*Ailurus fulgens fulgens* and *A. f. styani*). Z Sarwar*, **KM Diamond**. Annual SICB meeting, Atlanta, GA
- Comparative jumping performance of the critically endangered dusty gopher frog (*Lithobates sevosus*). **KM Diamond**, C Alton, C Hanks*, P Chittoori**, R Yang*, S Poo. Annual SICB meeting, Atlanta, GA
- Vertebral variation in juvenile goby fish, 'o'opu nōpili, in relation to escape performance. E Pertuit*, M Maheu*, A Johnson*, RW Blob, HL Schoenfuss, **KM Diamond**. Annual SICB meeting, Atlanta, GA
- 2024 The effects of age on exploratory behavior in Endler's Guppies (*Poecilia wingei*). SE Heckmann*, **KM Diamond**. Annual SICB meeting, Seattle WA & Rhodes Symposium, Memphis TN
- The effects of body length on undisturbed swimming patterns of Endler's guppies (*Poecilia wingei*). Y Li*, **KM Diamond**. Annual SICB meeting & Rhodes Symposium, Memphis TN
- Influence of waterfall-climbing style and ontogeny on fast-start performance in gobiid fishes. S Ravichandran*, **KM Diamond**, AM Palecek-McClung, RW Blob. Annual SICB meeting, Seattle WA
- Grazing kinematics and the evolution of waterfall climbing in gobiid fishes. RW Blob, **KM Diamond**, Elpidio Buelecope‡, JA Cullen, T Maie, HL Schoenfuss. Annual SICB meeting, Seattle WA
- 2023 Starting a biological machine learning lab at a PUI. **KM Diamond**. Southeastern regional meeting for the Society for Integrative and Comparative Biology, Blacksburg, VA
- Feeding kinematics of algal grazing in the Gobiid fish *Sicydium punctatum*: a missing link in the evolution of goby climbing mechanics. RW Blob, **KM Diamond**, J Cullen, T Maie, HL Schoenfuss. International Congress of Vertebrate Morphology, Cairns, Australia
- Functional plasticity of the swim bladder in a vocal fish. L Rogers, N Lozier, Y Sapozhnikova, **KM Diamond**, J Sisneros. Annual SICB meeting, Austin, TX
- Allometry of size and performance in the desert horned lizard (*Phrynosoma platyrhinos*). **KM Diamond**, C Olson, KL Utsumi, MA Eifler, DA Eifler. Annual SICB meeting, Austin, TX
- Bigger, faster, stronger? The influence of body size on goby fast-start performance. S Ravichandran*, **KM Diamond**, HL Schoenfuss, RW Blob, AM Palecek-McClung. Annual SICB meeting, Austin, TX
- 2022 The effects of *plod2* on zebrafish 3D craniofacial phenotype. A Burtner*, D Siddiqui*, **KM Diamond**. University of Washington Undergraduate Research Symposium.
- Quantifying craniofacial phenotype of zebrafish with mutations in the *sost* gene, known homolog for human *SOST* associated with hyperostosis. SL Leake*, **KM Diamond**. University of Washington Undergraduate Research Symposium.

Using a semi-automated screening tool to quantify craniofacial variation in meox1 crispant zebrafish. K Alvarado*, **KM Diamond**. University of Washington Undergraduate Research Symposium.

- 2021 Computational anatomy and geometric shape analysis enables analysis of complex craniofacial phenotypes in zebrafish mutants. **KM Diamond**, RY Kwon, AM Maga. European Calcified Tissue Society meeting (virtual)

Machine learning-based segmentation and landmarking of 2D fish images. **KM Diamond**, BB Avants, AM Maga. Annual SICB meeting (virtual)

Geometric morphometrics of climbing kinematics in waterfall climbing goby fishes. JG Griner*, AM Palecek, **KM Diamond**, HL Schoenfuss, RW Blob. Annual SICB meeting (virtual)

Ontogenetic change in performance: do innovations constrain performance? HL Schoenfuss, **KM Diamond**, R Lagarde‡, T Maie, RW Blob. Annual SICB meeting (virtual)

Measuring craniofacial variability in zebrafish using computational anatomy. **Diamond KM**, Kwon RY, Maga AM. Presented at SCRI research symposia (virtual December 2020) and annual SICB meeting (virtual)

- 2020 Linking morphology, performance, and behavior in the migration of stream goby fishes. **KM Diamond**, JG Griner*, R Lagarde‡, D Ponton‡, KE Powder, HL Schoenfuss, JA Walker, RW Blob. Annual SICB meeting. Austin, TX.

Comparative body shapes of amphidromous goby fishes living in different predator regimes. JF Griner*, **KM Diamond**, RW Blob. Annual SICB meeting. Austin, TX.

Do predators take advantage of prey blind spots? In-stream analysis of predator-prey interactions in Hawaiian stream fishes. NG Schneider*, AJ McKamy*, **KM Diamond**, RW Blob. Annual SICB meeting. Austin, TX.

Locomotor mechanics of juvenile alligators reveals ontogenetic changes in the roles of the fore- and hindlimbs. M Iijima‡, VD Munteanu, **KM Diamond**, RW Blob. Annual SICB meeting. Austin, TX.

Humeral strains during climbing in green iguanas: testing biomechanical release as a mechanism promoting morphological transitions in arboreal vertebrates. VD Munteanu, **KM Diamond**, CJ Mayerl, RW Blob. Annual SICB meeting. Austin, TX.

- 2019 Morphology, performance, & anti-predator strategies of gobiid fishes across predator regimes. **KM Diamond**, R Lagarde‡, D Ponton‡, K Powder, HL Schoenfuss, RW Blob. ICVM. Prague, Czech Republic.

Climbing and escape performance over migration pulses in the Hawaiian goby. **KM Diamond**, HL Schoenfuss, RW Blob. Annual SICB meeting. Tampa, FL

Bendy to the bone: a study of climbing goby vertebrae. GK Forker*, **KM Diamond**, RW Blob. Focus on Creative Inquiry. Clemson, SC

- 2018 Do predators take advantage of prey blind spots? AJ McKamy*, NG Schneider*, **KM Diamond**, RW Blob. SE SICB meeting, Clemson, SC.

Bendy to the bone. GK Forker*, **KM Diamond**, RW Blob. SE SICB meeting, Clemson, SC.

Take it or leave it. Fast-start modulation in the great sculpin *Myoxocephalus polyacanthocephalus*. LM Penrod, A Daddino, **KM Diamond**, JL Johansen, JF Steffensen‡, P Domenici‡. Annual SICB meeting. San Francisco, CA

Where to find the best bugs: a study of habitat use among lizard species with different movement strategies. EA Mcalpine-Bellis*, GE Garrison, S Gilbert Smith*, JRV Klein*, KL Utsumi*, **KM Diamond**, D Eifler, M Eifler. Annual SICB meeting. San Francisco, CA

Does the form of stress matter? A comparison of Pacific sand lance (*Ammodytes hexapterus*). A Daddino, **KM Diamond**, L Penrod, JL Johansen, JF Steffensen‡, P Domenici‡. Annual SICB meeting. San Francisco, CA

Do the best come first? Locomotor performance over the course of migration events in the amphidromous goby fish, *Sicyopterus stimpsoni*. **KM Diamond**, HL Schoenfuss, RW Blob. Annual SICB meeting. San Francisco, CA

- 2017 Locomotor performance over the course of migration events in the amphidromous goby fish, *Sicyopterus stimpsoni*. **KM Diamond**, RW Blob. Southeastern SICB meeting. Blacksburg, VA

Comparative escape performance of prey fish species that face different degrees of range overlap with predators. **KM Diamond**, HL Schoenfuss, JA Walker, RW Blob. Clemson Biological Sciences Annual Student Symposium. Clemson, SC (3rd place oral presentation)

Does ontogenetic environment influence escape response? Comparative escape responses of goby fishes through ontogeny and among islands. **KM Diamond**, HL Schoenfuss, JA Walker, RW Blob. Annual SICB meeting. New Orleans, LA

Assessing the impacts of environmental contaminants on escape behavior in the migratory stream goby *Sicyopterus stimpsoni*. AM Rubin*, **KM Diamond**, HL Schoenfuss, RW Blob. Annual SICB meeting. New Orleans, LA & Clemson Biological Sciences Annual Student Symposium. Clemson, SC (2nd place in best poster)

Comparative waterfall climbing kinematics of goby fishes from Hawai'i and Réunion: are recently evolved behaviors less variable? RM Keefe*, **KM Diamond**, R Lagarde‡, D Ponton‡, RS Bertram, HL Schoenfuss, RW Blob. Annual SICB meeting. New Orleans, LA

Mechanical specializations of the fin rays in waterfall-climbing gobiid fishes. NK Taft, BN Taft, **KM Diamond**, HL Schoenfuss, RW Blob. Annual SICB meeting. New Orleans, LA

- 2016 Does ontogenetic environment correlate with escape behavior in fish? **KM Diamond**, HL Schoenfuss, JA Walker, and RW Blob. Southeastern SICB meeting. Durham, NC

Is fish escape performance influenced by flow? Tests in the Hawaiian stream goby, *Sicyopterus stimpsoni*. **KM Diamond**, HL Schoenfuss, JA Walker, and RW Blob. Annual SICB meeting. Portland, OR

Waterfall-climbing performance of gobiid fishes from La Réunion: how conservative are novel functional behaviors? RS Bertram, HL Schoenfuss, R Lagarde‡, D Ponton‡, **KM Diamond**, T Offerle, and RW Blob. Annual SICB meeting. Portland, OR

Field observation of intraspecific and predatory attack behaviors of the Hawaiian sleeper fish, *Eleotris sandwicensis*. AM Rubin*, **KM Diamond**, HL Schoenfuss, and RW Blob. Presented at Annual SICB meeting. Portland, OR & Clemson Biological Sciences Annual Student Symposium. Clemson, SC

- 2015 Impacts of flow speed and attack direction on the fast-start escape response of the Hawaiian goby, *Sicyopterus stimpsoni*. **KM Diamond**, HL Schoenfuss, JA Walker, and RW Blob. Southeastern SICB meeting. Atlanta, GA.

Just keep swimming! Testing effects of flow speed and stimulus direction on escape behavior in a Hawaiian gobioid. **KM Diamond**. Clemson Biological Sciences Annual Student Symposium. Clemson, SC

Fast-start escape behavior in juvenile Hawaiian gobies, *Sicyopterus stimpsoni*: testing effects of flow speed and stimulus direction. **KM Diamond**, HL Schoenfuss, JA Walker, and RW Blob. Presented at: Southeastern SICB meeting. Chapel Hill, NC (2014) and Annual SICB meeting. West Palm Beach, FL (2015)

- 2014 Sistemática y biogeografía de *Bothriechis schlegelii* (Berthold, 1846) (Serpentes: Viperidae), una aproximación molecular y morfológica. JP Hurtado Gómez‡, **KM Diamond**, CL Parkinson, and JM Daza Rojas‡. IV Congreso Colombiano de Zoología. Cartagena de Indias, Colombia

Combined morphological and molecular analysis of rattlesnake phylogeny. AM Fenwick, **KM Diamond**, CL Parkinson. Southwestern Association of Naturalists Annual Meeting. Stillwater, OK

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- 2013 Evolutionary history and polymorphism in the Eyelash Palm Pit-viper (*Bothriechis Schlegelii*): **KM Diamond**, and CL Parkinson. Presented at: Southeastern Ecology and Evolution Conference and Showcase of Undergraduate Research. Orlando, FL

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